

Changing an entity's name

A simple guide to update an entity's name with Script Extender.

Handling Display Names

Introduction

In this tutorial, we'll explore how to manage entity display names in Larian's game using `DisplayName.NameKey`. We'll discuss how to extract the display name handle, why using `DisplayName.NameKey` is superior to `Osi.SetStoryDisplayName`, and demonstrate functions to add and remove prefixes to display names dynamically.

Extracting `DisplayName.NameKey`

To manipulate an entity's display name, we need to extract the handle from `DisplayName.NameKey`. Here's how you can achieve this:

1. Retrieve the Entity Object:

```
1 | local entityObject = Ext.Entity.Get(entity)
```

2. Extract the Handle:

```
1 | local handle =  
    entityObject.DisplayName.NameKey.Handle.Handle
```

3. Get the Display Name:

```
1 | local displayName =  
    Ext.Loca.GetTranslatedString(handle)
```

Why Use `DisplayName.NameKey` ?

Using `DisplayName.NameKey` is better than `SetStoryDisplayName` because it provides a persistent and consistent way to handle display names. `SetStoryDisplayName` creates temporary mappings that are lost upon game reload, leading to issues where names may revert to "Not Found".

`DisplayName.NameKey` ensures that the changes remain consistent across game sessions.

Adding Prefix to Display Name

To add a prefix to an entity's display name, we can use the following function. It checks if the prefix already exists to avoid redundant updates:

```

1 | local function UpdateEntityDisplayName(entity,
2 | prefix)
3 |
4 |     local entityObject = Ext.Entity.Get(entity)
5 |     local handle =
6 |     entityObject.DisplayName.NameKey.Handle.Handle
7 |     --Extracting the handle
8 |     if handle then
9 |         local displayName =
10 | Ext.Loca.GetTranslatedString(handle)
11 |         if not displayName:find(prefix) then
12 |             local updatedName = prefix .. " "
               .. displayName

               Ext.Loca.UpdateTranslatedString(handle,
               updatedName)
               end
           end
       end
   end

```

Usage Example:

```

1 | UpdateEntityDisplayName(someEntity,
   | "Stoneskin")

```

This function ensures that the display name is updated only if the prefix is not already present.

Removing Prefix from Display Name

To remove a prefix from an entity's display name, we can use the following function:

```

1 | local function
2 | RemovePrefixFromDisplayName(entity, prefix)
3 |
4 |     local entityObject = Ext.Entity.Get(entity)
5 |     local handle =
6 |     entityObject.DisplayName.NameKey.Handle.Handle
7 |     -- Extracting the handle
8 |     if handle then
9 |         local displayName =
10 | Ext.Loca.GetTranslatedString(handle)
        local updatedName =
        displayName:gsub("^" .. prefix .. " ", "")
        Ext.Loca.UpdateTranslatedString(handle,
        updatedName)
        end
    end
end

```

Usage Example:

```

1 | RemovePrefixFromDisplayName(someEntity,
    "Stoneskin")

```

This function removes the specified prefix if it exists in the display name.

Practical Application

Updating Display Name on Turn Start

To update the display name based on active status during the entity's turn start:

```

1 | local statusToPrefixMap = {
2 |     STONESKIN = "Stoneskin",
3 |     GOLDEN = "Golden",
4 |     -- Add other statuses and their
5 |     corresponding prefixes here
6 | }
7 |
8 | Ext.Osiris.RegisterListener("TurnStarted", 1,
9 | "after", function(object)
10 |

```

```

11 |
12 |     -- Check and update display name based on
13 |     status
14 |     for status, prefix in
15 |     pairs(statusToPrefixMap) do
16 |         if HasActiveStatus(object, status) == 1
17 |     then
18 |         UpdateEntityDisplayName(object,
19 |         prefix)
20 |     end
21 |     end
22 |
23 |     -- Add your existing TurnStarted logic here
24 | end)

```

Removing Prefix on Death

To remove the prefix when the entity dies:

```

1 | Ext.Osiris.RegisterListener("Dying", 1,
2 | "after", function(entity)
3 |
4 |     -- Check and remove prefix based on status
5 |     for status, prefix in
6 |     pairs(statusToPrefixMap) do
7 |         RemovePrefixFromDisplayName(entity,
8 |         prefix)
9 |     end
10 |
11 |     -- Add your existing Dying logic here
12 | end)

```

Conclusion

Using `DisplayName.NameKey` for managing entity display names provides a robust solution for maintaining consistency across game sessions. The functions for adding and removing prefixes ensure that display names are updated dynamically based on the entity's status. This approach prevents issues related to temporary mappings and provides a better overall experience in managing display names.